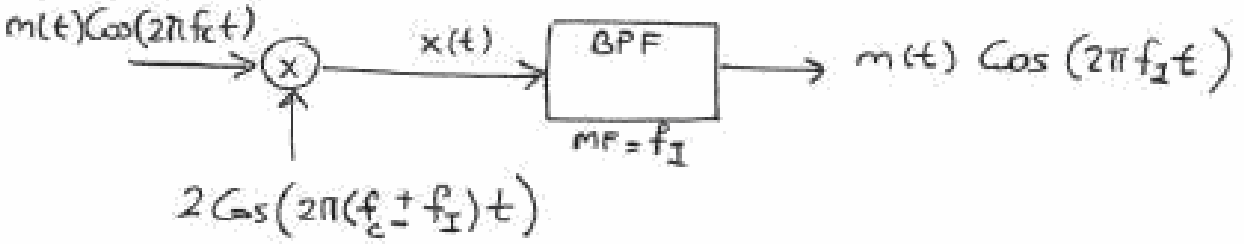
Ex. (Lathi) Frequency Mixer (Convertor) (Frekans Karıştırıcı)

Changes carrier freq. f_c to f_I

$$m(t)\cos(2\pi f_c t) \longrightarrow m(t)\cos(2\pi f_I t)$$



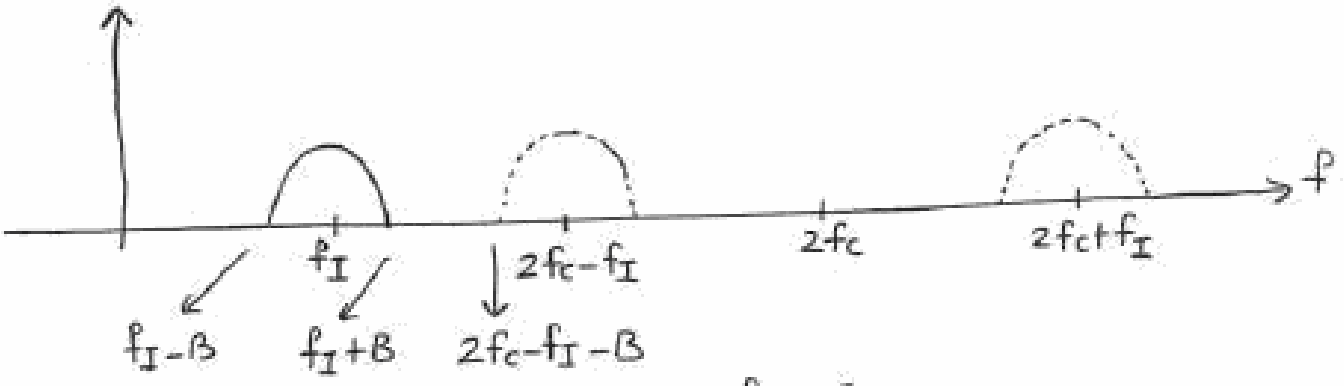
$$x(t) = 2\,m(t)\cos(2\pi f_c t)\cos(2\pi f_k t), \qquad f_k = \begin{matrix} f_c + f_I \\ f_c - f_I \end{matrix} \text{ veya}$$

$$= m(t)\left[\cos(2\pi(f_c - f_k)t) + \cos(2\pi(f_c + f_k)t)\right]$$

$f_k = f_c - f_I \Rightarrow x(t) = m(t)\left[\cos(2\pi f_I t) + \cos(2\pi(2f_c - f_I)t)\right]$

$f_k = f_c + f_I \Rightarrow x(t) = m(t)\left[\cos(2\pi f_I t) + \cos(2\pi(2f_c + f_I)t)\right]$

↙ BGF'den geçmez.



$f_I - B \geq 0$

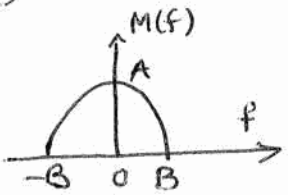
$\boxed{f_I \geq B}$

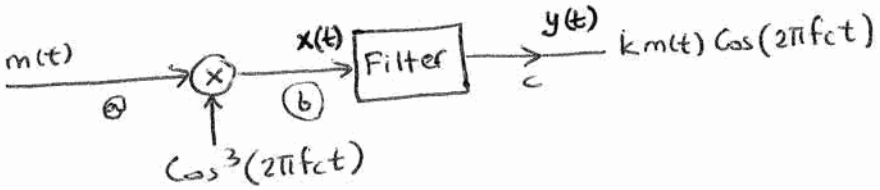
$2f_c - f_I - B \geq f_I + B$

$2(f_c - f_I) \geq 2B \Rightarrow$ $\boxed{f_c - f_I \geq B}$

Ex. (Lathi, Prob. 4.2.4)

You are asked to design a DSB-SC modulator to generate a modulated signal $k\,m(t)\cos(2\pi f_c t)$, where $m(t)$ is a signal band-limited to B Hz. Figure shows a DSB-SC modulator available in the stock room. The carrier generator available generates not $\cos(2\pi f_c t)$, but $\cos^3(2\pi f_c t)$. Explain whether you would be able to generate the desired signal using only this equipment. You may use any kind of filter you like.





a) What kind of filter is required?

b) Determine the signal spectra at points b and c, and indicate the frequency bands occupied by this spectra.

c) What is the min. usable value of f_c?

d) Would this scheme work if the carrier generator output were $\cos^2(2\pi f_c t)$? Explain.

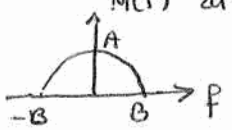
e) Would this scheme work if the carrier " " " $\cos^n(2\pi f_c t)$ for any integer $n \geq 2$?

$\cos^3\alpha = \cos\alpha\cos^2\alpha = \cos\alpha\,\frac{1}{2}[1+\cos 2\alpha] = \frac{1}{2}\cos\alpha + \frac{1}{2}\cos\alpha\cdot\cos 2\alpha = \frac{1}{2}\cos\alpha + \frac{1}{4}[\cos\alpha + \cos 3\alpha] = \frac{1}{4}[3\cos\alpha + \cos 3\alpha]$

$m(t)\cos^3(2\pi f_c t) = m(t)\,\frac{1}{4}\left[3\cos(2\pi f_c t) + \cos(2\pi\, 3 f_c t)\right] \longrightarrow \boxed{\text{BPF}}_{CF=f_c} \rightarrow \frac{3}{4}m(t)\cos(2\pi f_c t)$

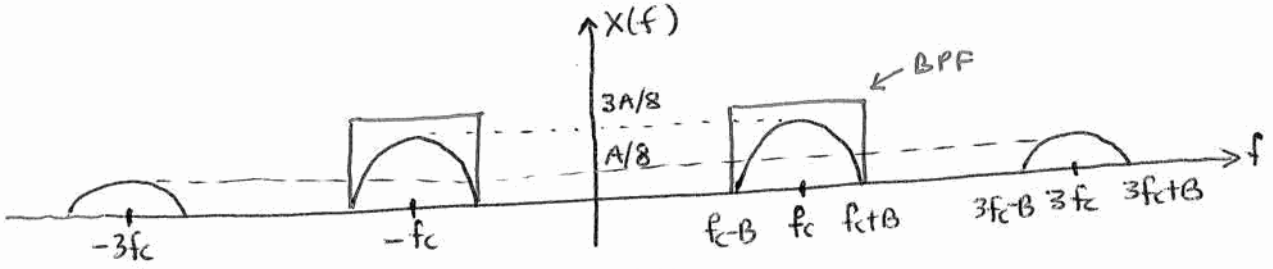
a) Filter = BPF with CF = f_c, BW = 2B Hz

b) M(f) at ⓐ

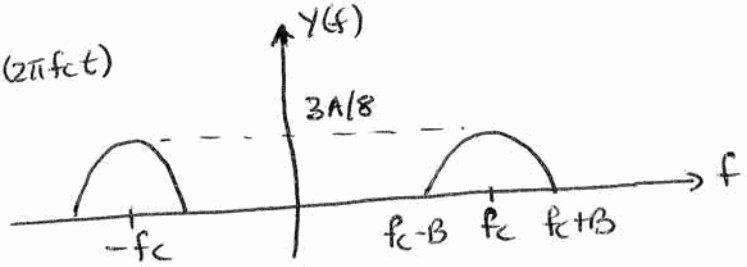


at ⓑ $x(t) = \frac{3}{4}m(t)\cos(2\pi f_c t) + \frac{1}{4}m(t)\cos(2\pi\, 3 f_c t)$

$X(f) = \frac{3}{8}\left[M(f-f_c) + M(f+f_c)\right] + \frac{1}{8}\left[M(f-3f_c) + M(f+3f_c)\right]$



at ⓒ $y(t) = \frac{3}{4}m(t)\cos(2\pi f_c t)$



c) $f_c - B \geq 0 \Rightarrow f_c \geq B$

$3f_c - B \geq f_c + B \Rightarrow f_c \geq B$

$\Rightarrow \boxed{f_c \geq B}$

d) $\cos^2(2\pi f_c t) = \frac{1}{2}\left[1 + \cos(2\pi\, 2 f_c t)\right]$ No

e) $\cos^n(2\pi f_c t)$ Yes if n is an odd number

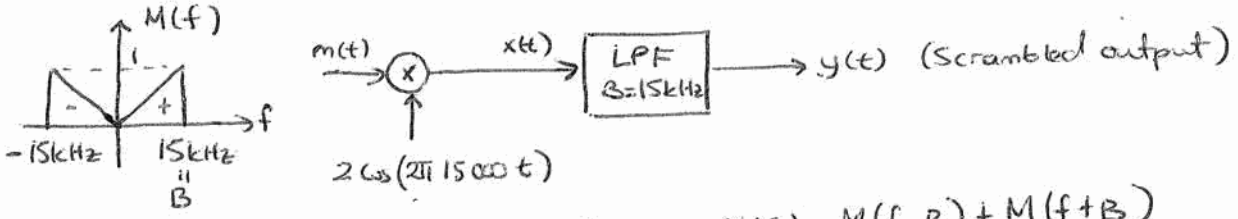
Ex. (Lathi, Pr. 4.2-9)

System shown in Figure is used for scrambling audio signals. The output $y(t)$ is the scrambled version of the input $m(t)$.

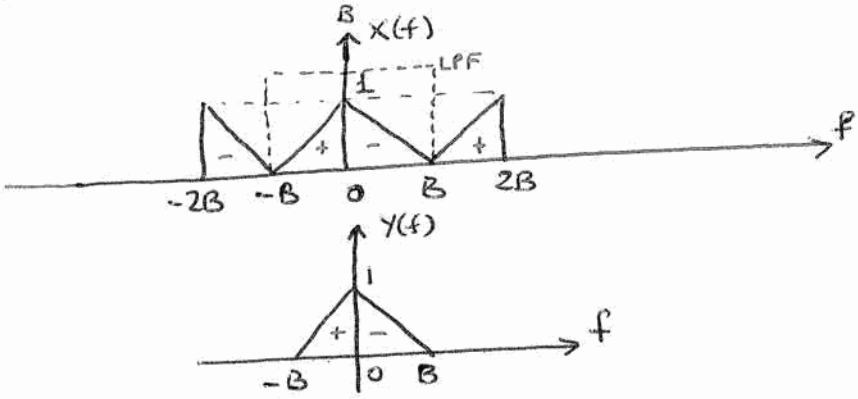
a) Find the spectrum of the scrambled signal $y(t)$.

b) Suggest a method of descrambling $y(t)$ to obtain $m(t)$.

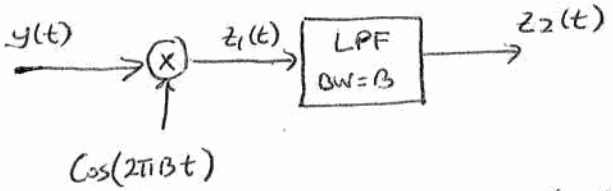
A slightly modified version of this scrambler was first used commercially on the 25-mile radio-telephone circuit connecting LA and Santa Catalina island.



a) $x(t) = 2m(t)\cos(2\pi\ 15000\,t) \rightarrow X(f) = M(f-B) + M(f+B)$



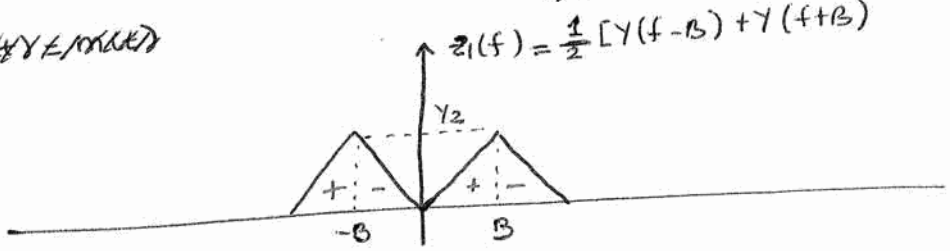
b)



$z_1(t) = y(t)\cos(2\pi Bt) = 2m(t)\cos^2(2\pi Bt) = m(t) + m(t)\cos(2\pi(2B)t)$

$z_1(t) \neq m(t)$

$z_1(f) = \frac{1}{2}[Y(f-B) + Y(f+B)]$



$z_2(f) = \frac{1}{2}M(f)$

